Chao-Yuan Chiang 姜兆源 (Tera)

B.S. Student (4th year) · National Sun Yat-sen University

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Research Interests: Robot manipulation and control · System identification · Sim-to-real transfer · Learning-based policies with control

Education

National Sun Yat-sen University (NSYSU) · B.S. (in progress) — Current

Department of Mechanical and Electro-Mechanical Engineering · fourth year

Nanyang Technological University (NTU), Singapore · Exchange student — Jan. – May 2027

Courses in smart manufacturing and robotics

Experience

UROP Research Intern · City Science Lab @ Taipei Tech · National Taipei University of Technology — Jul. – Aug. 2026

- Learned robot-arm fundamentals in MuJoCo: workspace, dynamics, and impedance torque compensation.
- Tried system identification on a physical robot arm.
- Worked in a three-person team on bimanual object handover with OpenArm: VR teleoperation demos, data augmentation with Isaac Lab Mimic, SmolVLA fine-tuning, and evaluation in simulation and on the real robot.

Research & Projects

Smart Waste-Sorting Bin with Multimodal Fusion and Fallback AI Decision-Making — NSTC project: Jul. 2026 – Feb. 2027

NSTC College Student Research Project · Advisor: Assoc. Prof. Cheng-Jung Yang

Sorts waste by fusing images with impact sounds and defers uncertain items to a cloud vision-language model; with cloud review, test accuracy reaches 99.49%. **My role:** Team lead; mechatronic integration and communication among the PC, Raspberry Pi, and ESP32. Teammates led the mechanism and the core AI model.

Bimanual Object Handover on OpenArm: Real2Sim2Real Imitation Learning — Jul.–Aug. 2026

Summer UROP research internship · City Science Lab @ Taipei Tech

Trained a policy for the OpenArm bimanual robot to hand over a Pringles can: VR teleoperation demos, data augmentation with Isaac Lab Mimic, SmolVLA fine-tuning, and evaluation in simulation and on the real robot. **My role:** Member of a three-person team (summer UROP research internship).

Ball-Shooting Robot for TEL Robot Combat 2025 — Jul.–Dec. 2025

TEL Robot Combat 2025 · Advisors: Assoc. Prof. Cheng-Jung Yang and Asst. Prof. Kai-Yi Weng

A ball-shooting robot that aims automatically at a nine-hole target; our team received the Creative Technology Award (Excellence). **My role:** Team captain; hardware architecture and mechanism design of the whole robot. Teammates handled software and AI.

Robot Navigation Course Projects

Robot Navigation course, National Tsing Hua University (NTHU) AI Alliance

Implemented A* and RRT* path planning and Stanley and LQR trajectory tracking, and practiced node communication in ROS 2. **My role:** Course projects.

Publications

Modality-Robust Confidence-Gated Cascade for Edge–Cloud Waste Sorting: Design and Evaluation on a Physical Prototype — Under review

Journal manuscript · IEEE Access (Regular Manuscript)

Development of an AIoT Smart Waste-Sorting Bin Integrating Multimodal Cross-Attention Fusion and an Evidential Cloud-Edge Fallback Decision Mechanism — Accepted with minor revisions

Conference paper · 29th Conference of the Chinese Society of Mechanism and Machine Theory (CSMMT 2026), Paper No. 22, National Taiwan University of Science and Technology, Taipei, Taiwan

Development of an AIoT Smart Waste-Sorting Bin Integrating Multimodal Cross-Attention Fusion and an Evidential Cloud-Edge Fallback Decision Mechanism — 2025–26

Undergraduate thesis · Department of Mechanical and Electro-Mechanical Engineering, NSYSU (in Chinese)

Awards & Grants

Creative Technology Award (Excellence) · TEL Robot Combat 2025 (Tokyo Electron), 10th edition — 2025

NSTC College Student Research Project (approved) · National Science and Technology Council (NSTC), Grant No. 115-2813-C-110-129-E — 2026